

TASK11: #51633

- i. Launch a AWS instances (OS can choose by your choice)
- ii. Host WordPress files under a user directory
- iii. Create an RDS instance
- iv. Connect RDS Database with your WordPress and run webpage using URL myawshosting.net

NB: Only Minimum access is granted

i. Launch the EC2 Instance

1. EC2 Console → Launch Instance
 - AML: **Amazon Linux 2023**
 - Instance type: t2.micro
 - Key pair: create/download .pem
 - Security Group (minimum access):
 - SSH (22) → My IP only
 - HTTP (80) → 0.0.0.0/0

2. Connect to the instance:

chmod 400 mykey.pem

```
ssh -i mykey.pem ec2-user@<EC2-PUBLIC-IP>
```

ii. Install httpd and Host WordPress Under a User Directory

bash

Update system

```
sudo dnf update -y
```

Install httpd, PHP, and required extensions

```
sudo dnf install -y httpd php php-mysqlnd php-fpm php-json php-gd php-mbstring php-xml wget unzip
```

Start and enable httpd

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

Download WordPress into ec2-user's home directory:

```
cd /home/ec2-user  
wget https://wordpress.org/latest.tar.gz  
tar -xzf latest.tar.gz  
sudo chown -R ec2-user:ec2-user /home/ec2-user/wordpress
```

Point httpd's DocumentRoot to this directory — edit:

bash

```
sudo nano /etc/httpd/conf/httpd.conf
```

Change:

```
DocumentRoot "/var/www/html"
```

to:

```
DocumentRoot "/home/ec2-user/wordpress"
```

Update the matching <Directory> block:

apache

```
<Directory "/home/ec2-user/wordpress">  
    AllowOverride All  
    Require all granted  
</Directory>
```

Allow httpd to traverse into the home directory + fix permissions:

```
sudo chmod 711 /home/ec2-user
```

SELinux context fix (required — Amazon Linux enforces SELinux by default)

```
sudo dnf install -y policycoreutils-python-utils  
sudo semanage fcontext -a -t httpd_sys_content_t "/home/ec2-user/wordpress(/.*)?"  
sudo restorecon -Rv /home/ec2-user/wordpress
```

- `semanage fcontext -a -t httpd_sys_content_t ".."` — registers a rule telling SELinux this path should be treated as web content httpd is allowed to read.
- `restorecon -Rv` — actually applies that label to the real files on disk (recursively, with verbose output).

If WordPress needs to **write** (uploads, plugin/theme installs), give that specific folder a writable label:

```
sudo semanage fcontext -a -t httpd_sys_rw_content_t "/home/ec2-user/wordpress/wp-content/uploads(/.*)?"
```

```
sudo restorecon -Rv /home/ec2-user/wordpress/wp-content/uploads
```

Restart httpd:

```
sudo systemctl restart httpd
```

Verify SELinux labels applied correctly

```
ls -Z /home/ec2-user/wordpress
```

iii. Create the RDS Instance

Console → RDS → Create database:

- Engine: **MySQL 8.0**
- Template: Free tier
- DB instance identifier: wordpress-db
- Master username / password: set and save these
- Instance class: db.t3.micro
- Storage: 20 GB gp2
- VPC: same VPC as EC2
- New security group: rds-sg
- Initial database name: wordpress_db

Minimum access rule on rds-sg:

- Inbound: MySQL/Aurora (3306) → source = **EC2's security group** (not an open CIDR)

Note the RDS **Endpoint** once the instance status is "Available" (found under Connectivity & security tab).

iv. Connect RDS to WordPress and Run the Site

1. Configure wp-config.php

```
cd /home/ec2-user/wordpress  
cp wp-config-sample.php wp-config.php  
vi wp-config.php
```

Set the four DB values to match RDS exactly:

```
define( 'DB_NAME', 'wordpress_db' );  
  
define( 'DB_USER', 'admin' );  
  
define( 'DB_PASSWORD', 'yourpassword' );  
  
define( 'DB_HOST', 'wordpress-db.xxxxxxxxxxxx.ap-south-1.rds.amazonaws.com' );
```

2. Install a MySQL/MariaDB client to test connectivity from EC2

On Amazon Linux 2023, mysql and mysql-client packages are no longer available — install the MariaDB client instead:

```
sudo dnf install -y mariadb
```

3. Test the connection

```
mysql -h wordpress-db.xxxxxxxxxxxx.ap-south-1.rds.amazonaws.com -u admin -p
```

Once connected:

```
SHOW DATABASES;  
  
If wordpress_db does not appear, create it manually (use underscore, not hyphen):  
  
sql  
  
CREATE DATABASE wordpress_db;  
  
SHOW DATABASES;  
  
EXIT;
```

4. Restart httpd and load the site

```
sudo systemctl restart httpd
```

Visit in browser:

<http://<EC2-PUBLIC-IP>>

This should load the WordPress installation wizard. Complete Site Title, Admin username/password, and email, then click **Install WordPress**.

Log in at:

<http://<EC2-PUBLIC-IP>/wp-admin>

Troubleshooting Reference

Symptom	Cause / Fix
"Error establishing a database connection"	DB_NAME/DB_USER/DB_PASSWORD/DB_HOST mismatch with RDS, or RDS security group not allowing EC2's SG on 3306
403 Forbidden	SELinux context not applied — re-run semanage fcontext + restorecon
Blank white page	PHP fatal error — check <code>sudo tail -30 /var/log/httpd/error_log</code>

Useful diagnostic commands

```
sudo systemctl status httpd  
  
sudo tail -30 /var/log/httpd/error_log
```

Check PHP MySQL extension is installed

```
php -m | grep -i mysql  
  
getsebool httpd_can_network_connect_db  
  
sudo setsebool -P httpd_can_network_connect_db 1
```

Check current wp-config.php DB values

```
grep DB_ /home/ec2-user/wordpress/wp-config.php
```

Option B — Local testing only (no real domain needed)

Edit the **hosts file** on your local Windows machine:

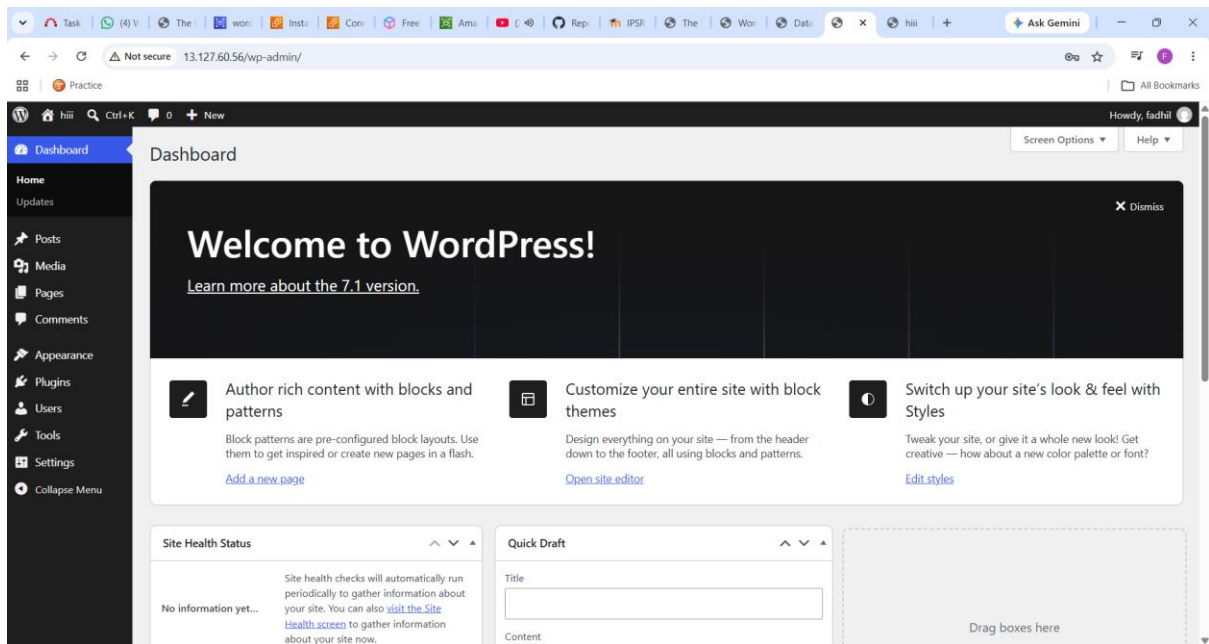
C:\Windows\System32\drivers\etc\hosts

Open Notepad **as Administrator**, open that file (select "All Files"), and add:

<EC2-or-Elastic-IP> myawshosting.net

OUTPUTS:

CALLED USING IP:



UDSING DOMIAN myawshosting:

